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High lithium oxide prevalence in the lithium solid–electrolyte interphase for high Coulombic efficiency

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Current electrolyte design for Li metal anodes emphasizes fluorination as the pre-eminent guiding principle for high Coulombic efficiency (CE) based largely on perceived benefits of LiF in the solid–electrolyte interphase (SEI). However, the lack of experimental techniques that accurately quantify SEI compositional breakdown impedes rigorous scrutiny of other potentially key phases. Here we demonstrate a quantitative titration approach to reveal Li_2O content in cycled Li anodes, enabling this previously titration-silent phase to be compared statistically with a wide range of other leading SEI constituents including LiF. Across diverse electrolytes, CE correlates most strongly with Li_2O above other phases, reaching its highest values when Li_2O particles order along the SEI, demonstrating integrated chemical–structural function. The beneficial role of Li_2O was exploited to create entirely fluorine-free electrolytes that breach >99% CE, highlighting electrolyte/SEI oxygenation as an underexplored design strategy.

Replacing the graphite anode in Li-ion batteries with Li metal is one of the most compelling strategies to meet targets of >750 Wh l^{-1} for electric vehicles[1,2], but Li cannot yet achieve Coulombic efficiencies (CEs) > 99.95% required for >1,000 cycle life[3–5]. During cycling, active Li inventory is lost through several pathways: first, as solubilized material at ultralow CE (< ~10%); second, via the evolution of electronically isolated Li^0 at intermediate CE (~ 10–95%); and third, through the irreversible formation of the solid–electrolyte interphase (SEI), which dominates losses as CE reaches 99% (refs. 5–7). To support design of very high-CE electrolytes, it is becoming crucial to understand the ideal target SEI composition and gain precise control over SEI formation so that only the most beneficial phases are formed.

Even after extensive research, SEI chemical and functional understanding is remarkably incomplete and reliant upon qualitative models[8,9]. Broadly, it is accepted that the SEI comprises inorganic and organic phases, with the former including lithium fluoride (LiF), lithium oxide (Li_2O), lithium carbonate (Li_2CO_3) and lithium nitride (Li_3N) depending on salt; and the latter including semi-carbonates, alkoxides (ROLi) and poly/oligomers depending on solvent[10]. More accurately determining SEI composition remains a major challenge due to its exceedingly low amounts (sub-μmol_{Li} cm^{-2} per cycle at >99% CE), instability and susceptibility to contamination during analysis, and limitations of spectroscopy techniques. Much understanding derives from X-ray photoelectron spectroscopy[9], which has low depth sensitivity (~10 nm), resulting in widespread use of sputter depth profiling that can create compositional artefacts such as fictitious LiF enrichment[11–13] from beam-induced decomposition of electrolyte salts and fragments.

Despite this, LiF has been proposed as a leading SEI descriptor, resulting in fluorination becoming the principal electrolyte design strategy (Supplementary Fig. 1)[14–17]. On the basis of bulk properties, LiF has been rationalized to be a desirable SEI phase given its chemical inertness, mechanical strength, low electronic conductivity and high interfacial energy[16]. Yet it is unclear whether bulk properties are relevant in the nanoscopic, complex SEI, and LiF is among the more

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resistive phases for Li^+ transport[18–21]. Additionally, mounting evidence questions whether LiF has a clear chemical benefit in the SEI[16,21,22] or instead plays the role of an inert building block, motivating a deeper look into which phases provide the major SEI functionality.

Lithium oxide (Li_2O) is the second major ionic phase present in all model descriptions of the SEI[9,10]. In Li-ion batteries, Li_2O is the fully reduced form of typical carbonate solvents and is thus presumed to play a role in the passivation of graphite[23,24]. On Li, however, Li_2O has received less focus, especially compared with LiF. We previously observed that nanostructured all-Li_2O SEI grown on Li foil possesses an approximately two times higher Li^+ conductivity than all-LiF SEI, thus supporting comparatively more homogeneous Li^+ flux, and rendering Li_2O more beneficial to SEI transport[18]. This hypothesis is consistent with observed O enrichment of many high-CE SEI[25,26] and in some advanced electrolytes[27,28]. Weighing the relative significance of Li_2O, LiF and other phases is however challenging without an ability to accurately determine their proportions throughout the SEI. Titration-based analysis of cycled electrode materials has emerged in recent years as a powerful technique to quantify inactive Li^0 and a growing number of SEI phases[7,29–32], but the lack of a selective technique appropriate for cycled Li anodes has left Li_2O as a titration-silent phase. Salt-derived phases such as Li_3N, S- and B-containing phases have also not yet been studied in titration frameworks.

To fill this gap, this work reports an alcohol-based titration followed by Karl Fischer analysis that selectively quantifies total Li_2O in cycled Li anodes. The method is integrated into a broader workflow designed to quantify other key SEI phases in parallel. These include LiF; Li_3N; S-, P-, B-containing phases; Li_2CO_3/semi-carbonates (collectively, $ROCO_2Li$); Li_2C_2; RLi and inactive Li^0, providing a methodology to probe the rich SEI composition and resolve key building blocks in the limit of high CE. Across ten diverse electrolytes spanning broad composition and CE range, a major fraction of capacity loss is allocated, substantially exceeding prior quantification benchmarks[6]. Contrary to conventional understanding, Li_2O is the most consistently abundant phase at high CE and the strongest CE descriptor, surpassing LiF, even in highly fluorinated electrolytes. Cryogenic high resolution transmission electron microscopy (Cryo-HRTEM) characterization further reveals that the distribution of Li_2O within the SEI affects CE, providing both chemical and morphological function. Leveraging these findings, we demonstrate the possibility to achieve highly competitive >99% Li CE using oxygenated, rather than fluorinated, solvents and salts. The results indicate that LiF enrichment is not strictly requisite for high-Li CE and highlight SEI oxygenation as a compelling but underexplored pathway to expand versatility of electrolyte design.

Quantitative titration of Li_2O in cycled anodes

Samples for titration analysis were generated over several plating/stripping cycles of Li on Cu to accumulate irreversible materials (herein termed SEI/Li^0 residuals), ending with a full stripping step. The Li_2O titration (Fig. 1a) is a two-step reaction in which residuals on Cu are reacted with 2-butoxyethanol to form LiOH from any present Li_2O ($Li_2O + BuOC_2H_4OH \rightarrow LiOH + BuOC_2H_4OLi$)[33,34]. The LiOH solution is injected into a Karl Fischer (KF) titrator and reacted to completion to produce H_2O (Supplementary Fig. 2), which is related back to the original amount of SEI Li_2O by charge balance (2 e^- detected at the KF electrode per unit LiOH = unit Li_2O; Fig. 1b and Methods). The 2-butoxyethanol/KF titration series is selective to Li_2O and LiOH over other SEI phases such as Li_2CO_3, Li_3N, LiH, LiF and metallic Li^0 (Fig. 1c). The latter reacts with 2-butoxyethanol to form an alkoxide rather than LiOH ($Li^0 + BuOEtOH \rightarrow BuOEtOLi + 0.5\ H_2$, as verified in Fig. 1c and Supplementary Fig. 3), thus is invisible to KF titration[34]. In this regard, using alcohol as the titrant provides an important advantage over acid-base titration of Li_2O[35], because aqueous titrations yield LiOH from both Li_2O and Li^0, which cannot be differentiated (Supplementary Note 1). LiOH was confirmed to be negligible in the initial SEI using infrared spectroscopy and low-water-content electrolytes (Supplementary Figs. 4 and 5 and Supplementary Note 2), and thus was produced exclusively upon butoxyethanol reaction with SEI Li_2O.

The Li_2O titration was integrated into a parallelized workflow to quantify capacity loss partitioning in a first exemplar electrolyte, 1 M $LiPF_6$ in ethylene carbonate (EC)/diethyl carbonate (DEC) (50/50 vol%), examined previously by titration but with no information on Li_2O[6]. A three-pronged titration scheme was adopted wherein batches of Cu/Li cells were cycled galvanostatically to a targeted capacity loss and diverted to either HCl, water or 2-butoxyethanol titration (Supplementary Figs. 6–8). Following prior work[6], the first two are used to quantify inactive Li^0, $ROCO_2Li$, Li_2C_2, RLi (HCl) and LiF, P-containing phases (H_2O). Samples used for those titrations were rinsed with anhydrous dimethyl carbonate (DMC) to remove electrolyte, which was confirmed to not alter the detected SEI composition or amount of Li^0 (Supplementary Fig. 9). The targeted capacity loss was ~1 mAh for acid/water and ~2 mAh for butoxyethanol, determined by minimum sample requirements for limiting phases in each method (Supplementary Notes 3 and 4 and Supplementary Fig. 10).

Titration results (Fig. 1d) were normalized to each cell's capacity loss and averaged across replicates, yielding the capacity loss partitioning (%) of the electrolyte shown in Fig. 1e (Supplementary Note 5). As observed previously[6], a major capacity loss mechanism in 1 M $LiPF_6$ EC/DEC forms inactive Li^0 (46.1% of total loss). However, among the remaining 53.9% comprising SEI phases, strong Li_2O prevalence could now be confirmed (yellow bars). Moreover, Li_2O was the most abundant detectable SEI phase (14.6% of total loss or 27.1% of SEI Li^+), followed more remotely by $ROCO_2Li$ (5.4% or 10.0%, respectively) and yet more minor phases: RLi, Li_2C_2, P-containing phases and LiF, each comprising ≤1% of total capacity loss. The remaining unidentified SEI losses (~32%) are shown in grey and correspond to solvent-derived phases (for example, poly/oligomeric species, polycarbonates or alkoxides), salt fragments besides LiF, or soluble phases; the former two currently elude analysis because they do not produce unambiguously measurable phases upon titration. To better understand the relationship between SEI composition and performance, the workflow is next applied to a broader series of electrolytes.

Li_2O and SEI quantification across diverse electrolytes

A range of electrolyte compositions, spanning conventional carbonate/ether classes with diverse salts and localized high-concentration electrolytes (LHCE), were selected to bridge low to high CE (Fig. 2a and Table 1). To account for different salt products in the SEI/Li^0 residuals (Fig. 2b), S- and B-containing phases were measured by inductively coupled plasma-atomic emission spectroscopy (ICP-AES), and Li_3N was analysed by a salicylate assay method[36] (Supplementary Fig. 11) in addition to the above-noted workflow after cells were cycled to their target capacity loss over 1–10 cycles (Supplementary Figs. 12–14). Fig. 2c shows the compositional breakdowns by rank in select electrolytes, with the complete data for all electrolytes in Fig. 2d (titration data: see Source Data, Supplementary Figs. 15–24, Supplementary Tables 1–9; SEI breakdown: Supplementary Fig. 25). Key findings in each electrolyte are first highlighted before a cross comparison by phase.

A first electrolyte examined was 1.37 M lithium bis(trifluoromethane)sulfonimide (LiTFSI) in 1,1,2,2-tetrafluoroethyl 2,2,3,3-tetrafluoropropyl ether (TTE)/DMC (70/30 vol%, CE = 40.4%), the composition of which was based on leading LHCEs[37,38], but replaces the usual lithium bis(fluorosulfonyl)imide (LiFSI) salt with LiTFSI. This electrolyte exhibited low CE consistent with inferiority of non-additive containing LiTFSI electrolytes compared with those based on LiFSI (Supplementary Fig. 1). Expectedly for very low CE, capacity loss was dominated by Li^0 (81.9% of total loss)[7]. The SEI comprised S-containing (S-cont.) phases from LiTFSI followed by minor

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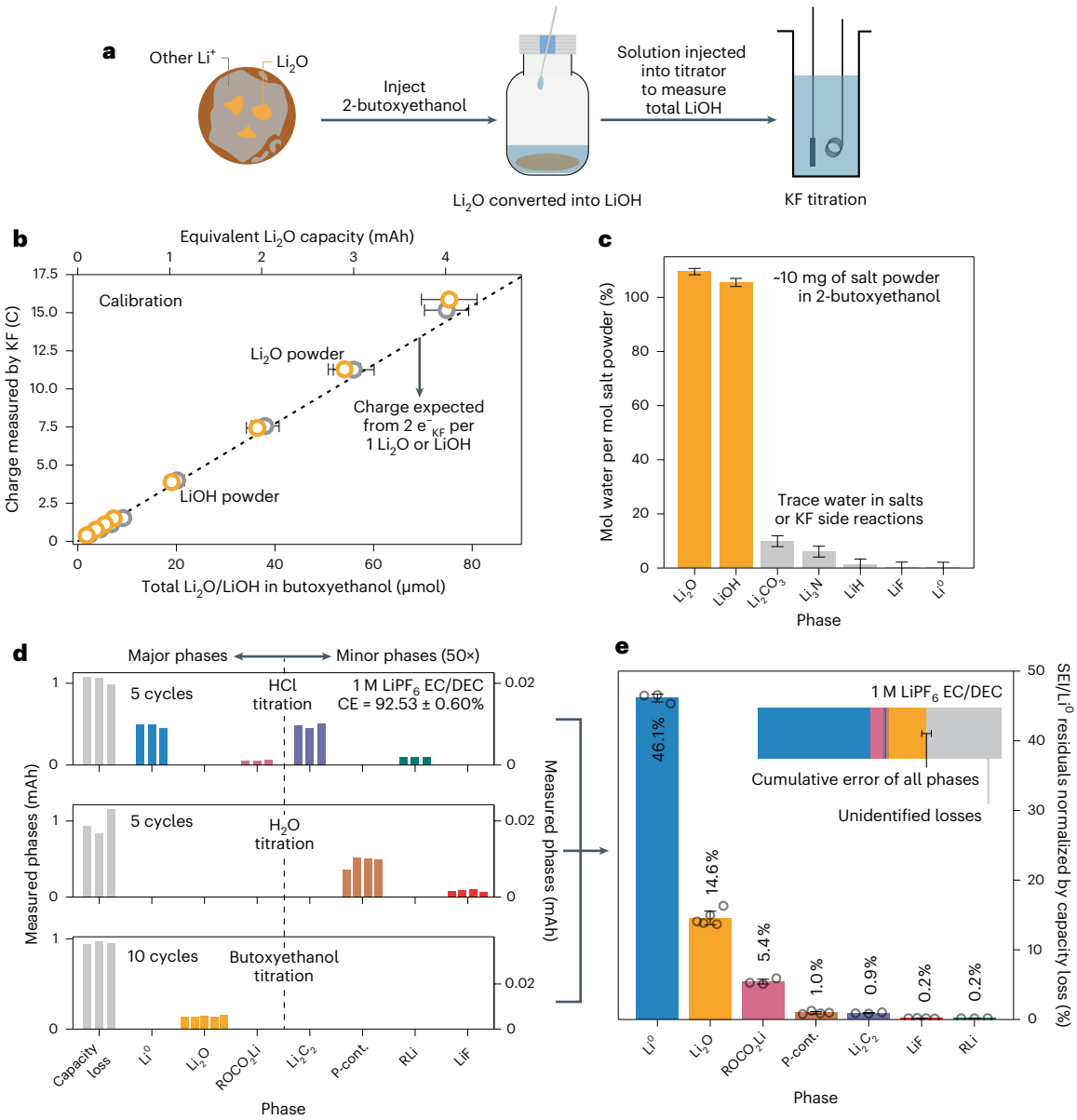

Fig. 1 | Titration strategy for quantification of Li_2O in cycled Li anodes. **a**, Illustration of the experimental workflow used for Li_2O quantification based on conversion of Li_2O into LiOH by reaction with 2-butoxyethanol, followed by titration of the resulting hydroxide using a coulometric KF method. **b**, Charge measured by the coulometric KF method after titration of standard solutions of LiOH and Li_2O in 2-butoxyethanol. The dashed line confirms the typical expected titrated 2 e^- charge for each unit Li_2O/LiOH. Slight deviation from the expected value at higher μmol values is due to residual water in the powders. **c**, Sensitivity of the KF method to other SEI phases and Li^0 in which 10 mg of each material was dissolved/reacted with 1 ml of 2-butoxyethanol and subsequently titrated by the KF method after 24 h. Error bars in **b** and **c** correspond to the error of the balance used to weigh Li_2O and LiOH samples (Supplementary Fig. 2) and not from statistics over multiple samples. **d**, Capacity loss in the 1 M $LiPF_6$ EC/DEC electrolyte attributed to individual phases, measured after 5–10 full cycles (0.5 mA cm^{-2}, 1.5 mAh cm^{-2}), including Li_2O in addition to Li^0, $ROCO_2Li$, Li_2C_2, RLi, LiF and P-containing (P-cont.) phases. **e**, Average partitioning of total capacity loss for each phase in which the absolute measured quantities of each phase in **d** are normalized by each cell's measured capacity loss. Data presented as mean ± standard deviation, and open markers denote individual data points ($n = 3$ for Li^0, $ROCO_2Li$, Li_2C_2, RLi; $n = 4$ for P, LiF; $n = 5$ for Li_2O). The inset indicates a cumulative representation of the average of each SEI phase and Li^0 from **d**.

amounts of other SEI phases, such as Li_2O and LiF. Organofluorine ($R\text{-}CF_x$) species, consistent with $TFSI^-$/TTE decomposition, were also observed by ^{19}F-NMR (Supplementary Fig. 26; Supplementary Figs. 27–35 for other electrolytes), though could not be quantified due to unknown fluorine stoichiometry x. Upon changing the solvent (1 M LiTFSI in 1,3-dioxolane (DOL)/1,2-dimethoxyethane (DME)), CE increased (59.8%), Li^0 decreased and Li_2O increased, though was still relatively minor. Whereas S- and F-containing phases in this electrolyte arise unambiguously from salt decomposition, Li_2O could also form from solvent reduction. Notably, this electrolyte yields the largest proportion of unresolved phases (grey region, 44.8% of total capacity loss) consistent with a poly/oligomeric-rich SEI[39]. A third electrolyte (1.37 M LiFSI 5TTE/5DMC) replaces LiTFSI with LiFSI but uses a blend of 5TTE/5DMC with suboptimal contact-ion pairing due to excess DMC compared with a high-CE LHCE analogue[38]. Still, it achieves higher CE (76.8%) and a higher SEI proportion of S-containing phases (35.8% of capacity loss) from participation of FSI^- in SEI formation. Li_2O and LiF are present in lower amounts than S-containing phases, indicating that FSI^- fragments in the SEI do not undergo complete reduction at these cycle conditions.